

WASP-104b

R-2414

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_190217 · created 2026-08-21T23:00:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458531.905 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.119 +/- 0.04
Transit depth (Rp/Rs)^2	1.42 +/- 0.96 [%]
Orbital Inclination (inc)	80.47 +/- 2.88
Airmass coefficient 1 (a1)	0.92 +/- 0.017
Airmass coefficient 2 (a2)	0.06 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.87 %
Best Comparison Star	#2 - [361, 282]
Optimal Aperture	2.81
Optimal Annulus	9.95
Transit Duration (day)	0.018 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.119 ± 0.04 vs 0.1214 (deviation 0.05σ)
Duration fit vs published	0.018 d vs 0.0742 d (deviation 2.78σ)
Mid-time O–C	-4.6 min
Depth SNR	2.7
Residual scatter	1.87 %

Light curve

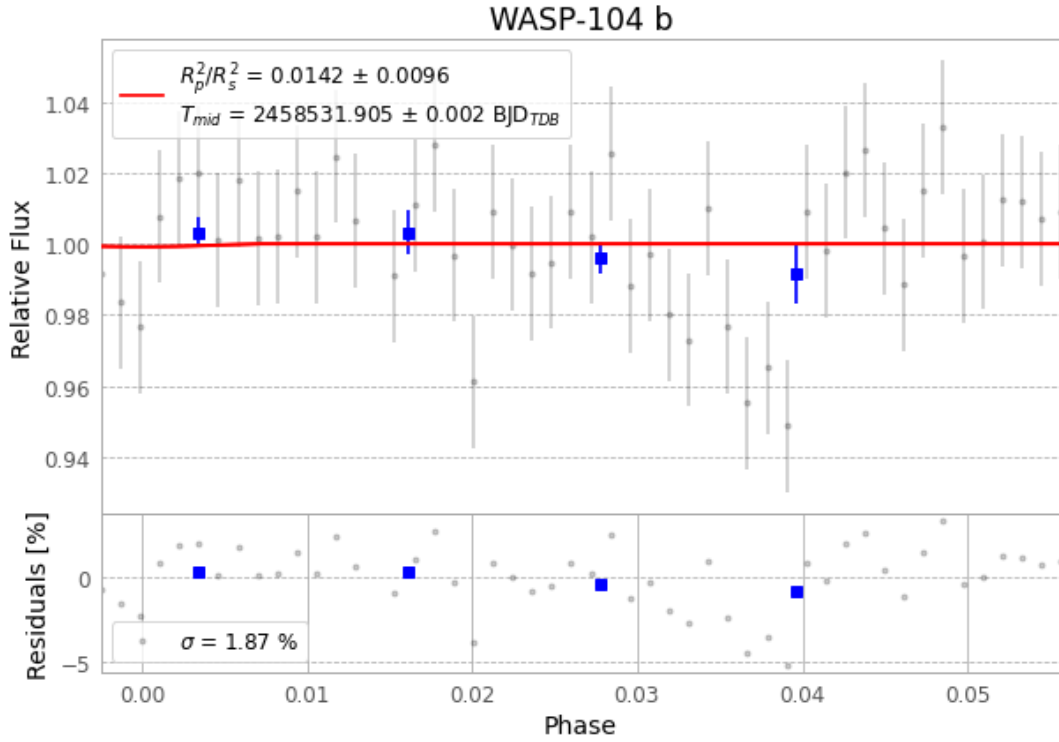


Figure 1 — FinalLightCurve_WASP-104 b_2019-02-17.png (SHA-256 04c9ae9bc3618793...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ce2e53721c00d2c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), WASP-104190217092713.FITS → WASP-104190217115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-190217.log (57f69b888b2a...), FinalLightCurve_WASP-104 b_2019-02-17.png (04c9ae9bc361...), FinalLightCurve_WASP-104 b_2019-02-17.csv (1b65608c8671...)

Compute resources

Wall time 97.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:00Z.